

BR Poll Aggregator - State Level Study

Luis Felipe Lins

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1 Introduction

In this study, we will test a poll aggregator for Brazilian presidential elections using state-level voting intentions. Using poll data from 11 different institutes¹ for presidential voting intentions *within states*, we aggregate these state-level voting intentions to a national level tracker. Then, we compare the performance of this state-level tracker to the traditional national-level trackers for the 2014, 2018, and 2022 elections. We find that the state-level tracker was considerably better than the national tracker in 2022, had a similar performance in 2018, and was worse in 2014. The explanation is open to debate. Based on the evidence available, we believe the main determinant is having a sufficiently high coverage of the national population in state-surveys.

Finally, we apply our method to the 2026 elections. The descriptive statistics suggest that 2026 is by far the year with more available and fresh polls, suggesting a relatively high degree of confidence in our estimates. Unlike previous years, in 2026 both the national and the state-level trackers suggest very similar results, with Lula and Flávio sharing roughly 50% of the national valid votes each.

2 Method

2.1 Poll Aggregation

We construct two national-level trackers: a *national-poll tracker*, based directly on nationwide surveys, and a *state-level aggregate*, constructed from a collection of state-level trackers. Because each state-level tracker is estimated using exactly the same procedure as the national-poll tracker, we begin by describing this common aggregation method.

For each election, we restrict the sample to presidential runoff polls and retain only surveys reporting preferences for both candidates. Poll results are normalized to a two-candidate vote-share scale, so that the reported shares of the two candidates sum to one. This makes polls directly comparable even when the original survey also reports undecided, blank, or null responses.

¹MDA, Quaest, Datafolha, AtlasIntel, Paraná Pesquisas, Nexus/FSB, PoderData, Ideia Big Data, Ipec/Ibope, Real Time Big Data, Futura/100 Cidades

The tracker is constructed at a daily frequency. For each day t , we consider all eligible polls released during the preceding 30 calendar days, including day t . Our baseline specification assigns greater weight to more recent polls. A poll released d days before t receives weight

$$w_{i,t} = \frac{1}{d_{i,t} + 1},$$

where $d_{i,t}$ is the number of days elapsed between poll i 's release and day t . Thus, a poll released on the same day receives weight 1, a one-day-old poll receives weight $1/2$, and so forth.

The estimate for candidate c on day t is therefore

$$\hat{p}_{c,t} = \frac{\sum_{i \in \mathcal{P}_t} w_{i,t} p_{i,c}}{\sum_{i \in \mathcal{P}_t} w_{i,t}},$$

where $\mathcal{P}_t$ denotes the set of polls released between $t - 29$ and t , and $p_{i,c}$ is candidate c 's normalized vote share in poll i .

We construct the tracker from September 1 through the date of the runoff election. Polls conducted during August are nevertheless retained whenever they fall inside the initial 30-day windows, preventing the tracker from starting with an artificially small information set.

This procedure is applied independently to nationwide polls and to polls conducted within each state. Consequently, the national-poll tracker and each state tracker differ only in the geographic population represented by the underlying surveys; the temporal aggregation method itself is identical.

2.2 From State Trackers to a National Aggregate

We next combine the individual state trackers into a single national estimate. The main challenge is that state-level polling coverage is incomplete: on a given date, some states may have one or more polls available within the 30-day window, while others may have no recent polling at all.

For states with available polling, we use the state tracker described above. For states without an estimate on a given date, our baseline specification imputes the state's vote share using the result of the previous presidential runoff. The mapping is made by political side rather than by candidate identity: the previous election's PT candidate is mapped to the current election's PT candidate, and the previous opponent is mapped to the current opponent. The imputed value remains fixed until the state obtains a polling-based estimate.

Let $\hat{p}_{s,c,t}$ denote the estimated vote share of candidate c in state s on date t , after filling states for which no current polling estimate is available. The national state-level aggregate is then

$$\hat{p}_{c,t}^S = \sum_{s \in \mathcal{S}} \omega_s \hat{p}_{s,c,t},$$

where $\mathcal{S}$ contains the 26 states, the Federal District, and overseas votes, and ω_s is state s 's share of valid votes in the previous presidential election,

$$\omega_s = \frac{V_{s,t-1}}{\sum_{j \in \mathcal{S}} V_{j,t-1}}.$$

Here, $V_{s,t-1}$ denotes the number of valid votes cast in geography s in the preceding presidential runoff. These weights are kept fixed throughout the tracking period.

Using previous-election vote totals rather than current-election turnout ensures that the aggregate can be constructed in real time, without using information that would only become available after the election. The resulting tracker can therefore be interpreted as the national vote share implied by the collection of state-level estimates available at each point in time.

3 Data

Our main source of data comes from Poder360’s dataset on polls². This dataset, however, does not contain poll data for 2026 nor for some reason for 2018. For these years, we reconstructed the dataset by filtering TSE’s database on voting polls³, where we are able to recover information on the poll’s registry number, coverage (national- or state-level), etc. Then, we asked ChatGPT to search the web for this database of surveys (300 in each year) collecting the actual figures in each year. This comes with a disclaimer: we checked a few numbers and they were correct, but errors might exist in these years.

4 Previous Election’s Results

4.1 2022 Election

Figure 1 shows the national-level (full blue line) and state-level (full red line) trackers. It shows that the state-level tracker was superior to the national tracker during all the considered time period, with absolute error below 1pp since September, 1st. The final error of the state-level tracker was 0.5pp versus 1.31pp from the national tracker. We can also compare these figures to other external trackers such as Globe Elections (0.4pp), The Economist (1.1pp), Veja (1.76pp), CNN Brasil (2.3pp), PollingData (0.46pp), Estadão (2.1pp)⁴. The median error was 1.28pp, with only 4 out of 14 institutes had error below our state-level aggregator.

For the 2022 elections, we had 69 national surveys from these 11 institutes. We also had 69 state-level surveys for 15 states, with the most surveys coming from SP (15), MG (8) and RJ (7). Despite the number of surveys being the same for state and national trackers, the composition was very different, with Ipec, Paraná and Quaest accounting for 71% of state-level surveys, versus 37% in national surveys. To rule out the explanation that the composition may explain the gap in national and state-level surveys, we add a dashed blue line where we only

²<https://basedosdados.org/dataset/fb38dbe8-03ce-46b4-a6b7-638ade03999c?table=b6df9e1c-cbcb-4dbd-893b-8645a51773e6>

³For 2026: <https://dadosabertos.tse.jus.br/pt/dataset/pesquisas-eleitorais-2026>. For 2018: <https://dadosabertos.tse.jus.br/pt/dataset/pesquisas-eleitorais-2018>.

⁴https://pt.wikipedia.org/wiki/Pesquisas_de_opinio_para_a_eleio_presidencial_no_Brasil_em_2022

consider national level polls with the institutes who did both state- and national-level polls: the *shared institutes* tracker. It does not seem superior to the unrestricted national level tracker.

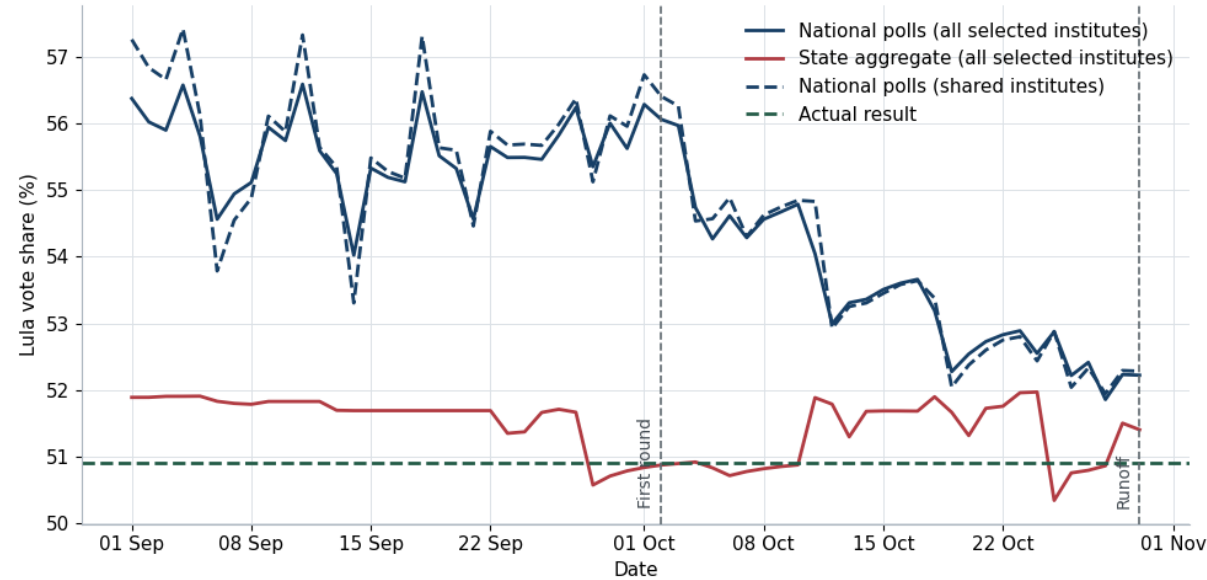


Figure 1: Trackers for 2022

Institute composition and accuracy (tracking period; errors are average absolute errors with equal survey weights):						
	National surveys	National share (%)	National error (pp)	State surveys	State share (%)	State error (pp)
Institute						
AtlasIntel/Internet	6	8.70	2.40	7	10.14	1.99
Datafolha	11	15.94	4.84	7	10.14	2.21
FSB	4	5.80	4.55	0	0.00	—
Futura	8	11.59	1.30	6	8.70	0.61
Ideia Big Data	2	2.90	3.27	0	0.00	—
Ipec	9	13.04	5.78	27	39.13	2.81
MDA	4	5.80	2.57	0	0.00	—
Paraná Pesquisas	8	11.59	1.79	12	17.39	1.42
PoderData	8	11.59	3.05	0	0.00	—
Quaest	9	13.04	3.63	10	14.49	4.91

Figure 2: Institutes for 2022

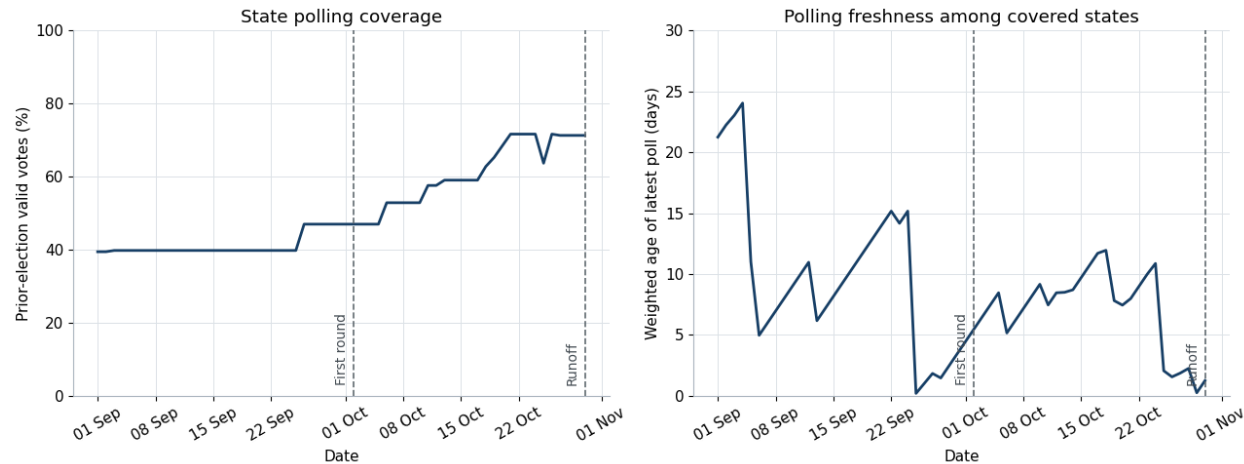


Figure 3: Coverage and Freshness for 2022

Now, we'd like to present our thesis on why our state-level tracker performed better than the national-level tracker in 2022. First, note that we already ruled out the possible composition effect, discussed in the previous paragraph. Another possible explanation would be that state-level survey errors cancel out. Neither does follow from inertia (using previous year's voting shares to missing states), since this would give us a 6pp error in 2022. Figure 6 shows this seems to be the case in the beginning of the period when coverage and accuracy were not good enough, but does not seem to be the case in the end of the period; accuracy improves over time. Also, if we look at this same chart for 2014 in Figure 10, we see that if accuracy is not good enough, the opposite might happen: the errors accrue. The fact that errors cancelled out in the beginning of 2022 can be pure luck in this case. Looking at Table 2 the more likely explanation, to me, lies on a combination of high coverage, fresh state polls, and the fact that in 2022 the state-level polls were on average more accurate than national-level polls. Almost all institutes who did both national-level and state-level surveys had on average lower errors at a state-level than at a national level (Atlas 2.4 vs. 1.99; Datafolha 4.84 vs 2.21; Futura 1.3 vs 0.61; Ipec 5.78 vs 2.81; Paraná Pesquisas 1.79 vs. 1.42; Quaest 3.63 vs. 4.91). As we will see in the next subsection, the same combination of factors which we use to explain the good results from 2022 can be used to explain the poor performance of the state-level tracker in 2014, and the poor performance before the 1st round of 2018, and the improved performance between both rounds.

4.2 2014 and 2018 Elections

In 2014, the state-level tracker (Figure 7) was considerably worse than the national-level tracker (3.9pp versus ~ 0.01 pp in the end). We had 22 national surveys and 13 state surveys in only 6 states, but with much less variety of institutes (Datafolha did all but one state polls). Table 8 also shows that unlike in 2022, in 2014 the state-polls were less accurate than the national-level polls. Coverage hit a maximum of $\sim 50\%$ just before the first round and decreased to 20% before the run off, while freshness was constantly above 10 days (Figure 9). As a result, during most of the time, the state-level tracker was using outdated voting shares from 2010, thus missing by a large share. Important states such as SP and MG had not up to date information since October, 8th and September 6th, respectively. The same combination of characteristics which seem to explain the good performance in 2022 seem to explain the poor performance in 2014.

In 2018 we have an interesting case. This was one of the most unpredictable elections in the 2000s. In the first days of August and September it wasn't clear which candidates would go to the run off. More importantly, there was a debate whether Lula would be able to run. Therefore, there were no runoff polls in the Haddad vs. Bolsonaro scenario before the 1st round (see Figure 13), meaning our tracker, available in Figure 11, was basically mimicking the 2014 voting shares. After that, the runoff state-polls quickly increased coverage, hitting 60% just before the election with freshness below 5 days. Looking at Figure 15 we see that our tracker's performance improved a lot after the first round, when the runoff was defined and more state-level polls became available, being similarly good relative to the national-level trackers, even though with an opposite signal.

5 The 2026 Elections

In 2026 we have the year with the highest state-level poll coverage ever. We have 21 national-level polls and 65 state-level polls. Since September, 1st we already have an 80% coverage of the country's population, hitting virtually 100% since September, 10th. Despite the high coverage, the average poll freshness is around 5 days, always below 10 days. These information can be seen under Figure 4, while the composition of polls can be seen in Figure 16.

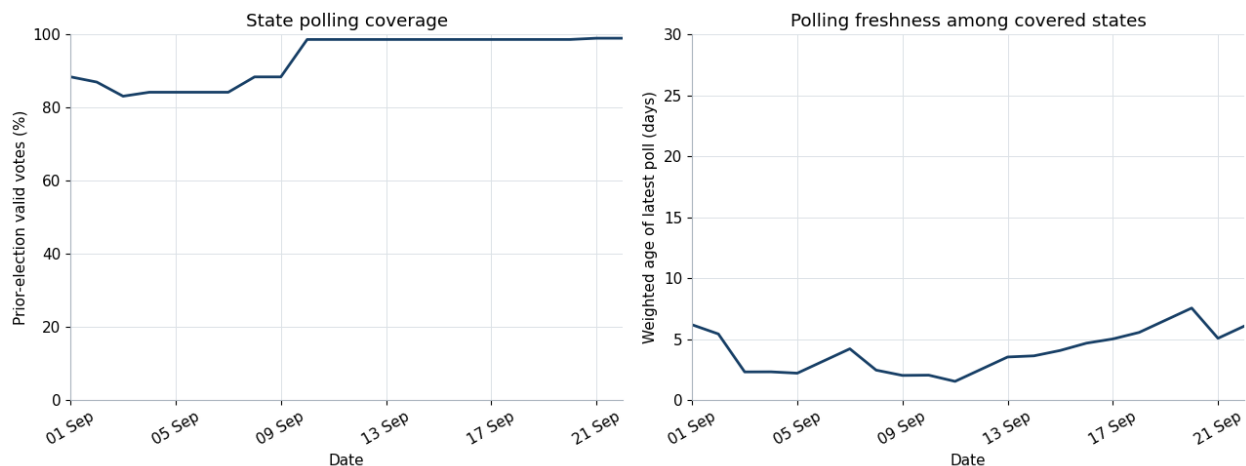


Figure 4: Coverage and Freshness for 2026

The actual trackers can be seen in Figure 5. Unlike in previous years, the national- and state-level trackers are very close to each other, all suggesting Lula and Flávio tied with 50% in the runoff, with Lula numerically ahead in both trackers by a tiny margin.

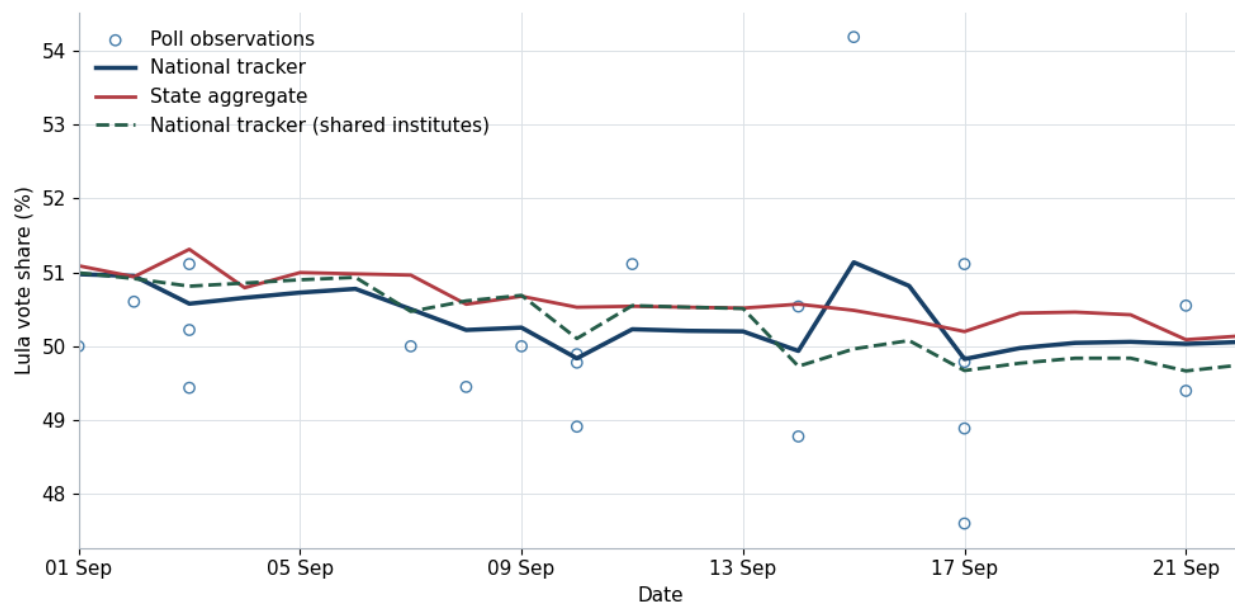


Figure 5: Trackers for 2026

A Appendix

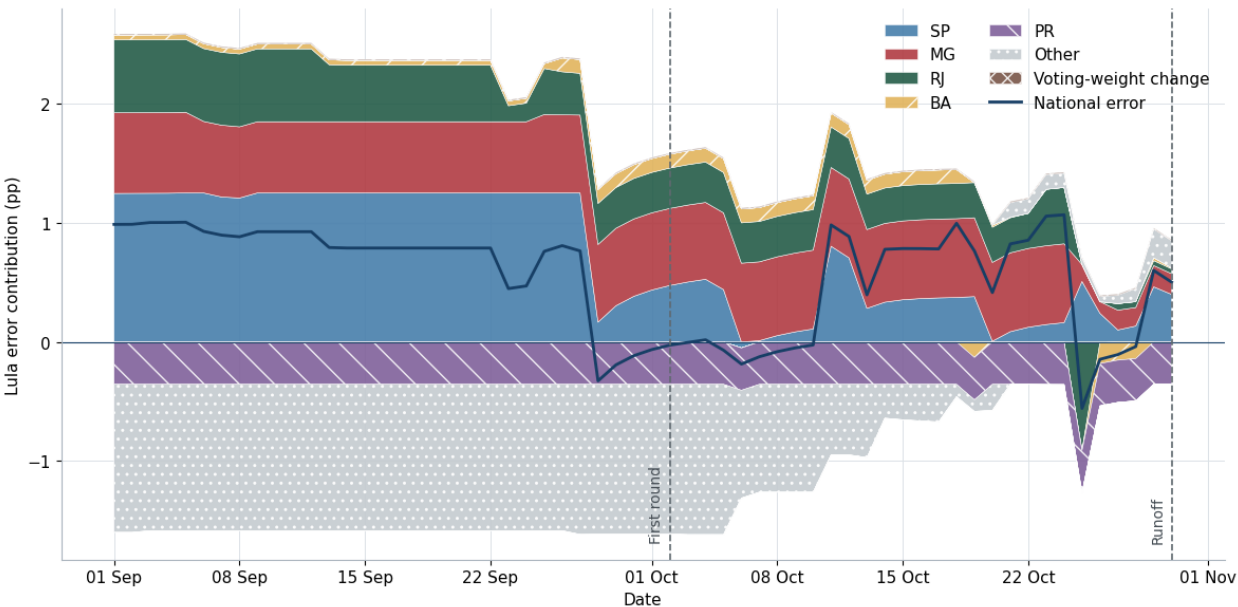


Figure 6: State Errors for 2022

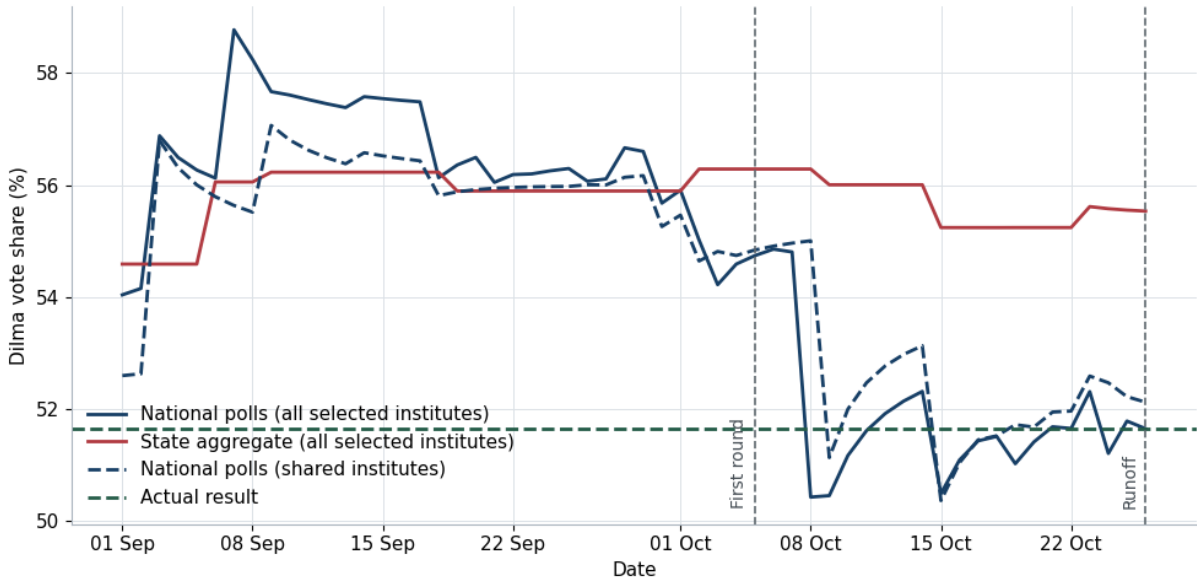


Figure 7: Trackers for 2014

Institute	National surveys	National share (%)	National error (pp)	State surveys	State share (%)	State error (pp)
Datafolha	15	68.18	2.76	12	92.31	4.98
FSB	0	0.00	—	1	7.69	6.78
MDA	6	27.27	3.67	0	0.00	—
Paraná Pesquisas	1	4.55	5.65	0	0.00	—

Figure 8: Institutes for 2014

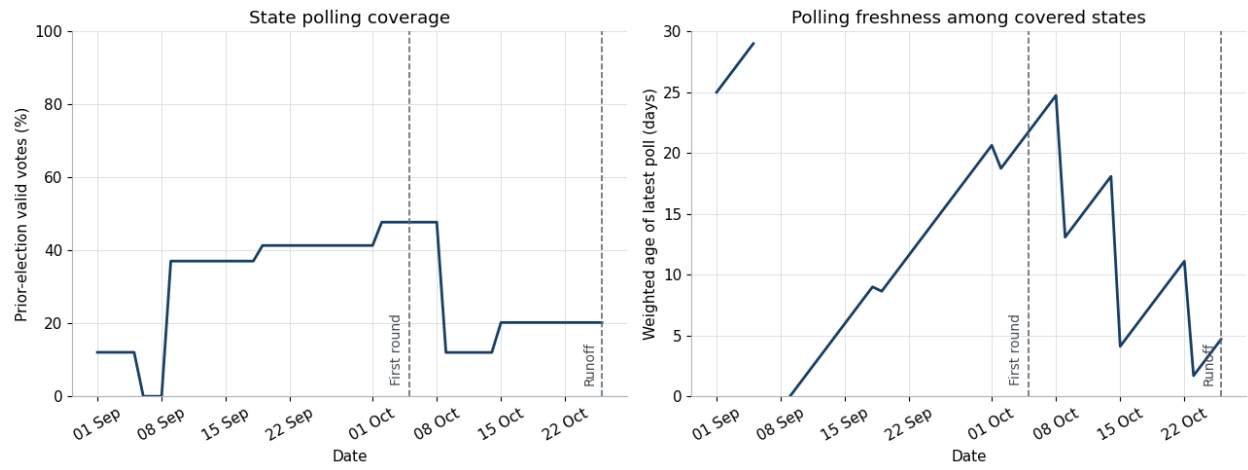


Figure 9: Coverage and Freshness for 2014

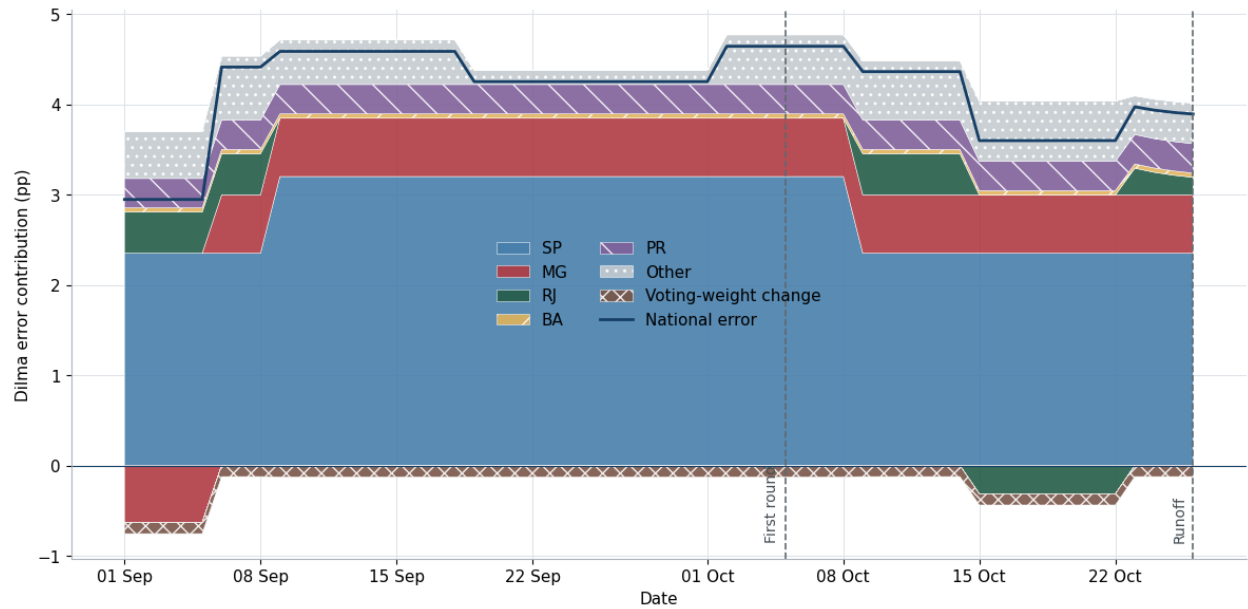


Figure 10: State Errors for 2014

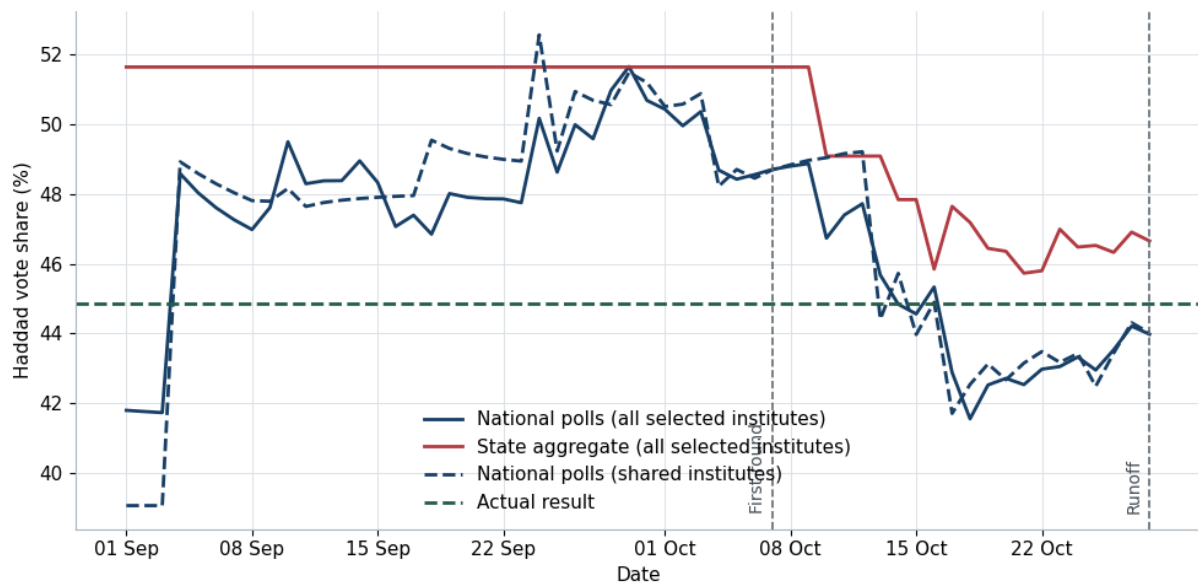


Figure 11: Trackers for 2018

Institute	National surveys	National share (%)	National error (pp)	State surveys	State share (%)	State error (pp)
Datafolha	12	24.49	3.84	0	0.00	—
FSB	5	10.20	3.29	0	0.00	—
Ideia Big Data	1	2.04	1.20	0	0.00	—
Ipec	11	22.45	4.56	29	76.32	4.76
MDA	5	10.20	3.24	0	0.00	—
Paraná Pesquisas	4	8.16	3.42	6	15.79	3.69
PoderData	2	4.08	6.52	0	0.00	—
Real Time Big Data	9	18.37	3.15	3	7.89	5.54

Figure 12: Institutes for 2018

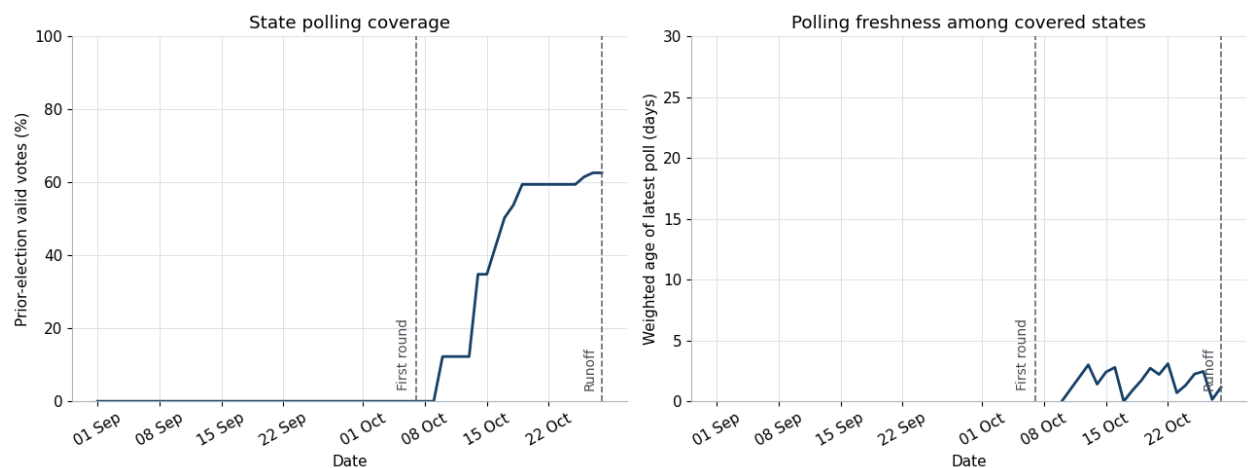


Figure 13: Coverage and Freshness for 2018

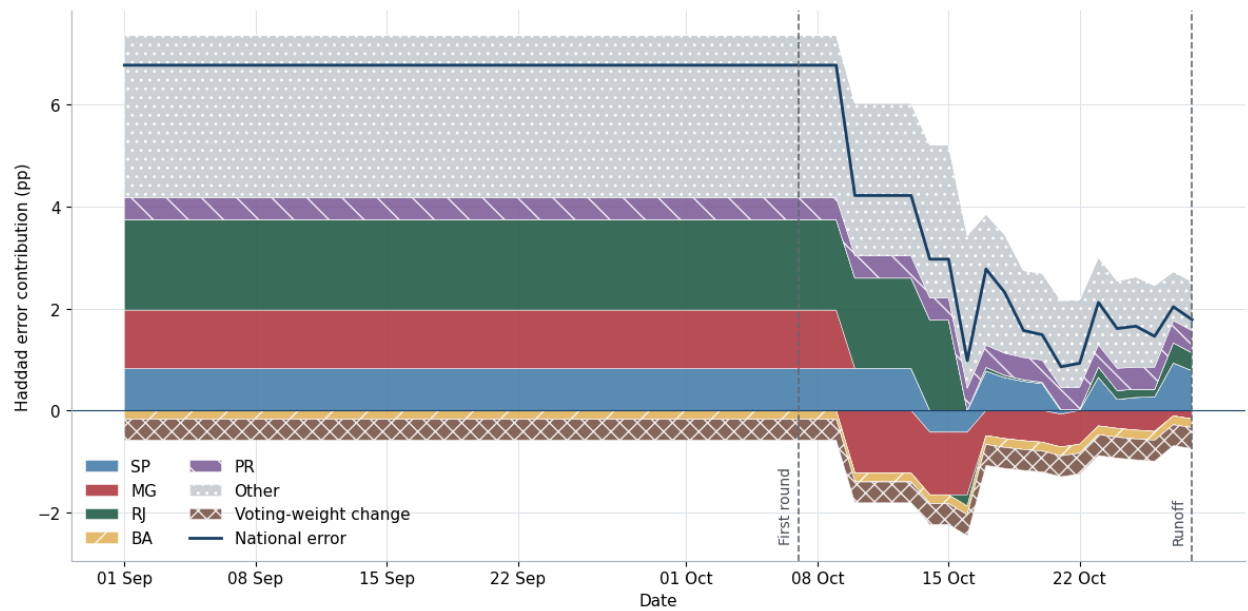


Figure 14: State Errors for 2018

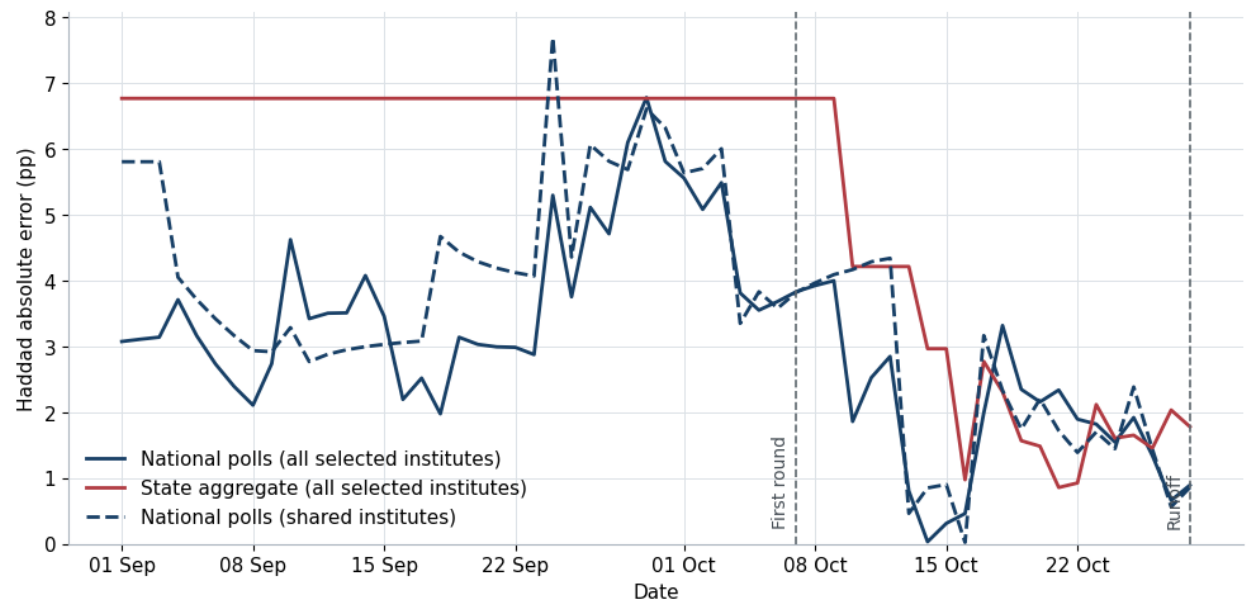


Figure 15: Tracker Errors for 2018

Institute	National surveys	National share (%)	State surveys	State share (%)
AtlasIntel/Internet	2	9.52	20	30.77
Datafolha	3	14.29	8	12.31
FSB	3	14.29	0	0.00
Futura	3	14.29	2	3.08
Ideia Big Data	1	4.76	0	0.00
MDA	1	4.76	0	0.00
Paraná Pesquisas	0	0.00	1	1.54
PoderData	3	14.29	0	0.00
Quaest	4	19.05	5	7.69
Real Time Big Data	1	4.76	29	44.62

Figure 16: Institutes for 2026